Statement of Teaching Philosophy

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Teaching Philosophy

I believe that the purpose of teaching is to provide students with a complete and in-depth understanding of the topics being studied, with the goal of preparing students both to apply the material and to build upon it. Achieving this requires a strong understanding of the theory behind the material in order to understand the strengths and limitations of the algorithms covered. This understanding is especially important in rapidly evolving fields such as robotics and AI, where students must be able to understand, evaluate, and adapt to new algorithms. In my own research, I frequently encounter problems for which existing algorithms are inadequate or whose assumptions do not hold. I therefore want students to understand not only how algorithms work, but also why they work, when they fail, and how to develop new approaches when existing methods are inadequate. This will allow students to extend and build upon the subject matter in their own work. Beyond this, students must understand the impact that their work will have on society to appropriately weigh the associated costs and benefits. This is especially important for fields such as robotics and AI, which will have a major impact on people's lives.

My teaching style combines Project-Based Learning with traditional lectures and coursework. I design my lessons to teach the fundamental concepts behind the material, while course projects require students to apply these concepts to real-world problems. I teach both the theoretical foundations of the material and its practical applications. I then assign projects and exam questions that require students to demonstrate their understanding of the theoretical background and how it influences practical applications. The projects are chosen based on requests from industrial collaborators or faculty members, which gives students the opportunity to create something that will be useful to our partners or will help faculty in their research. This also gives students an opportunity to collaborate with industrial partners, which may lead to employment opportunities, or with faculty members, which could lead to future research opportunities. My assignments consist of a set of challenging problems that require students to apply the subject matter to problems outside of those studied in class. These problems are designed to require students to think critically and reason about the ideas being studied.

Finally, I have a strong interest in expanding learning beyond the university setting and beyond traditional university students. I have found that many people are interested in robotics, but view it as being too esoteric and too complicated to understand. Many people also have misconceptions about what AI is and what AI and robotics are capable of. Robotics and AI will play a growing role in our society and it is important for people to have a basic understanding of them. It is important to provide people from other fields with the tools to apply AI and robotics to their own areas of expertise. These individuals may also bring valuable perspectives and insights that we have not considered.

Teaching Experience

I have served as the instructor of record for large core courses, such as the undergraduate course in Artificial Intelligence at Rutgers University (Introduction to Artificial Intelligence). This course enrolled more than 100 students and consisted of a combination of programming projects, exams, and written assignments. My goal for this course was to help students develop a deep understanding of AI principles that enables them not only to implement existing algorithms but also to recognize their assumptions, limitations, and opportunities for innovation. I structured my AI course into three major sections: a section on deterministic reasoning, a section on reasoning under uncertainty,

and a section on learning and reasoning in unknown environments. The section on deterministic reasoning covered search, local search (hill climbing), adversarial search, logic, and planning. The section on reasoning under uncertainty covered Bayesian networks, hidden Markov models, Kalman filters, particle filters, Markov decision processes, and POMDPs. The section on learning and reasoning in unknown environments covered decision trees, regression, expectation maximization, support vector machines, reinforcement learning, and neural networks. I assigned a project for each section, as well as a midterm and final exam. This syllabus is based loosely on the structure of the textbook *Artificial Intelligence: A Modern Approach* by Stuart Russell and Peter Norvig, which I used for this course. A copy of the syllabus I used for this course can be provided upon request.

During my time at Aalborg University, I was an instructor for the Reading in Robotics course at the Department of Materials and Production, which was a graduate-level course designed to introduce students to the research process. My goal for this course was to instill in students good research practices that they can carry into their own research, as well as to introduce students to the peer review and paper publication process. This course provided students with hands-on experience conducting literature reviews and introduced them to the submission, peer review, and publication process. Teaching at Aalborg University gave me a unique opportunity to experience how European universities structure their courses. In particular, Aalborg University is one of the pioneers in Project-Based Learning, where courses are centered on a semester-long project which requires students to apply the concepts from the course to solve a real-world problem. One such project developed the object detector that we used for grasping on the move, while another used visual servoing to do grasping.

Mentoring Students

Throughout my academic career, I have taught and mentored hundreds of undergraduates and about 20 graduate students. While at Texas A&M, I mentored numerous junior lab members, including members of the Motion Planning group. The students I mentored in this group went on to do research in the fundamentals of sampling-based motion planning. I also mentored students as part of Texas A&M's Research Experiences for Undergraduates (REU) program and was a judge for the REU poster session. At the University of Queensland, I served as the Head Tutor for the University's AI course and assisted numerous students in this role. At Rutgers, I taught hundreds of students while teaching the Introduction to AI course and mentored students at PRACSYS Lab. While at Aalborg University, I served as a mentor for a pair of undergraduates who were part of France's ASSE International Student Exchange Program. These students designed a specialized sampling-based planner for grasping objects. I also mentored two undergraduates who designed the base planner and navigational component for the mobile manipulator used to do grasping on the move. Many of the graduate students I have taught or mentored have gone on to successful careers in academia and industry, with some becoming faculty members. Similarly, many undergraduates I have taught or mentored have gone on to graduate study or successful careers in industry.

Future Teaching Interests

I am qualified to teach a broad range of undergraduate and graduate courses, including courses in artificial intelligence, robotics, computational geometry, machine learning, algorithms, and complexity. I am especially interested in teaching courses in artificial intelligence, robotics, and computational geometry. I would be particularly interested in creating courses on robotics and motion planning. In such a course, I would first cover some of the foundational papers in the field (e.g., configuration space, PRMs and RRTs, RRT*) and then some of the most important and interesting recent work (e.g., applications of machine learning to motion planning).